Chapter 6. Normed Spaces

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Overview

Results from Chapters 1–3 are cited as *Fact B.k*; they are stated in the separate *Background* notes. Several normed spaces have appeared so far; this chapter studies the basic properties of general normed spaces. The dual space — the space of bounded linear maps into the scalars — is essential for understanding a normed space, and we identify the duals of the ℓ^p-spaces and of Hilbert spaces. We also study elementary properties of bounded linear maps between Banach spaces (Baire category, open mapping, closed graph, uniform boundedness) and apply them to Fourier transforms.

6.1 Dual Spaces

Motivation. A normed space is often best understood through the continuous linear functionals on it: measures act on functions, sequences in ℓ^q act on ℓ^p. Identifying these dual spaces, and knowing there are enough functionals to separate points (Hahn–Banach), is the basic tool of functional analysis.

6.1.1 Bounded linear maps

Definition (operator norm)

For a linear map $T : X \to Y$ between normed spaces put

$$\|T\| = \sup\{\|T(x)\| : x \in X,\ \|x\| \le 1\}.$$

If $\|T\| < \infty$, T is a **bounded linear map** and $\|T\|$ is its **norm**.

The norm of T measures how much T stretches the unit ball $\{x \in X : \|x\| \le 1\}$ of X inside Y. For example, $z \mapsto \alpha z : \mathbb{C} \to \mathbb{C}$ has norm $|\alpha|$.

Exercise 6.1.1

Show that a linear map $T : X \to Y$ is bounded if and only if there is $M > 0$ with

$$x \in X \implies \|T(x)\| \le M\|x\|,$$

and that in this case $\|T\| \le M$.

Solution [Supplement]**.** If T is bounded, take $M = \|T\| + 1$ (or $\|T\|$ if positive): for $x \neq 0$, $x/\|x\|$ has norm 1, so $\|T(x)\| = \|x\|\,\|T(x/\|x\|)\| \le \|T\|\|x\|$; for $x = 0$ both sides vanish. Conversely, if $\|T(x)\| \le M\|x\|$ for all x, then $\|T(x)\| \le M$ whenever $\|x\| \le 1$, so $\|T\| \le M < \infty$. □

Exercise 6.1.2

Prove that $\|T\| = \sup\left\{\dfrac{\|T(x)\|}{\|x\|} : x \in X,\ x \neq 0\right\}$.

Solution [Supplement]**.** Let S be the right side. For $x \neq 0$, $\|T(x)\|/\|x\| = \|T(x/\|x\|)\| \le \|T\|$, so $S \le \|T\|$. Conversely, if $0 < \|x\| \le 1$ then $\|T(x)\| \le \|T(x)\|/\|x\| \le S$, and $T(0) = 0$; so $\|T\| \le S$. □

Exercise 6.1.3

Show that the set of all bounded linear maps between two normed spaces is a vector space, and that $T \mapsto \|T\|$ is a norm on it.

Solution [Supplement]. If S, T are bounded and $a \in \mathbb{C}$, then for $\|x\| \le 1$

$$\|(S+T)(x)\| \le \|S(x)\| + \|T(x)\| \le \|S\| + \|T\|, \qquad \|(aT)(x)\| = |a|\|T(x)\| \le |a|\|T\|,$$

so $S+T$, aT are bounded with $\|S+T\| \le \|S\| + \|T\|$ and $\|aT\| \le |a|\|T\|$; applying the latter to a^{-1} gives $\|aT\| = |a|\|T\|$ for $a \neq 0$. Clearly $\|T\| \ge 0$, and $\|T\| = 0$ forces $T(x) = 0$ for $\|x\| \le 1$, hence for all x by scaling, i.e. $T = 0$. □

Proposition 6.1.1

For a linear map $T : X \to Y$ between normed spaces, the following are equivalent:

(a) T is bounded;

(b) T is continuous;

(c) T is continuous at some point $x_0 \in X$.

Proof. The inequality $\|T(x) - T(y)\| \le \|T\|\|x - y\|$ gives (a)⇒(b), and (b)⇒(c) is trivial. If T is continuous at x_0, there is $\delta > 0$ with $\|x - x_0\| < \delta \Rightarrow \|T(x) - T(x_0)\| < \epsilon$ (take $\epsilon = 1$, say). By linearity this is the same as $\|y\| < \delta \Rightarrow \|T(y)\| < \epsilon$, from which $\|T\| \le \epsilon/\delta$. ■

Proposition 6.1.2

For normed spaces X, Y, the set $B(X,Y)$ of bounded linear maps from X to Y is a normed space. If Y is a Banach space, so is $B(X,Y)$.

Proof. That $B(X,Y)$ is a normed space is Exercise 6.1.3. Assume Y is a Banach space and let $\langle T_n \rangle$ be Cauchy in $B(X,Y)$. For each $x \in X$, $\|T_n(x) - T_m(x)\| \le \|T_n - T_m\|\|x\|$, so $\langle T_n(x) \rangle$ is Cauchy in Y and has a limit, which we call $T(x)$. This gives a map $x \mapsto T(x)$, and $T \in B(X,Y)$ with $\|T_n - T\| \to 0$ (Exercise 6.1.4). ■

Exercise 6.1.4

Complete the proof of Proposition 6.1.2.

Solution [Supplement]. T is linear as a pointwise limit of linear maps. Given $\epsilon > 0$ choose N with $\|T_n - T_m\| < \epsilon$ for $m, n \ge N$. For $\|x\| \le 1$ and $n, m \ge N$, $\|T_n(x) - T_m(x)\| < \epsilon$; letting $m \to \infty$ gives $\|T_n(x) - T(x)\| \le \epsilon$. Hence $T_N - T$ is bounded, so $T = T_N - (T_N - T) \in B(X,Y)$, and $\|T_n - T\| \le \epsilon$ for $n \ge N$. □

Definition (dual space)

A linear map with values in the scalars $\mathbb{C}$ is a **linear functional**. The normed space of all bounded linear functionals on X is denoted $X^* = B(X, \mathbb{C})$. By Proposition 6.1.2, X^* is always a Banach space; it is the **dual space** of X.

Example 1

$f \mapsto \int_0^1 f$ is a bounded linear functional on $(C[0,1], \|\cdot\|_{\sup})$. However

$$f \mapsto \int_{\mathbb{R}} f(x)\,dx \tag{1}$$

is not even a linear functional on $(C_0(\mathbb{R}), \|\cdot\|_{\sup})$, since the integral of $f \in C_0(\mathbb{R})$ need not be finite. Restricting the domain to $C_c(\mathbb{R})$ gives a linear functional which is not bounded: choosing $f_n \in C_c(\mathbb{R})$ with $0 \le f_n \le 1$ and $f_n = 1$ on $[-n, n]$ gives $\|f_n\|_{\sup} = 1$ but $\int_{\mathbb{R}} f_n \to \infty$. On the other hand, (1) is a bounded linear functional on $(L^1(\mathbb{R}), \|\cdot\|_1)$ by inequality Fact B.9 ($|\int f| \le \int |f|$).

Example 2

Let (X, μ) be a measure space, p, q conjugate exponents, and $g \in L^q(\mu)$ $(1 \le q \le \infty)$. Consider

$$\varphi_g : f \mapsto \int_X fg\,d\mu, \qquad f \in L^p(\mu).$$

By Hölder, $|\varphi_g(f)| \le \|g\|_q \|f\|_p$, so φ_g is a bounded linear functional on $L^p(\mu)$ with

$$\|\varphi_g\| \le \|g\|_q, \qquad g \in L^q(\mu).$$

In fact equality holds.
Case $p = \infty$, $q = 1$. For fixed $g \in L^1(\mu)$ we want $f \in L^\infty(\mu)$ with $\|f\|_\infty \|g\|_1 = |\int_X gf|$. Take

$$f(x) = \begin{cases} \dfrac{|g(x)|}{g(x)}, & g(x) \neq 0, \\ 0, & g(x) = 0; \end{cases}$$

this function is written $\operatorname{sgn} g$. Then $gf = |g|$, so $\int_X gf = \|g\|_1 = \|f\|_\infty \|g\|_1$ (if $g \neq 0$).
Case $p = 1$, $q = \infty$. Fix $g \in L^\infty(\mu)$ with $\|g\|_\infty \neq 0$. For small $\epsilon > 0$ the set $\{x \in X : |g(x)| > \|g\|_\infty - \epsilon\}$ has positive measure; choose a subset E of it with $0 < \mu(E) < \infty$ (possible if (X, μ) is σ-finite) and put $f = \chi_E \cdot \operatorname{sgn} g \in L^1(\mu)$. Then

$$\int_X gf = \int_E g \cdot \operatorname{sgn} g = \int_E |g| \ge \mu(E)(\|g\|_\infty - \epsilon) = \|f\|_1(\|g\|_\infty - \epsilon),$$

so $\|\varphi_g\| \ge \|g\|_\infty - \epsilon$, and hence $\|\varphi_g\| = \|g\|_\infty$.
Case $1 < q < \infty$. For fixed $g \in L^q(\mu)$ put $f = |g|^{q/p} \operatorname{sgn} g$. Then $|f|^p = |g|^q$, so $\|g\|_q^{q-1} = \|f\|_p$ (as $q/p = q - 1$), and $fg = |g|^q$. Hence

$$\int_X fg = \int |g|^q = \|g\|_q^q = \|g\|_q \|f\|_p,$$

and $\|\varphi_g\| = \|g\|_q$.

So for any measure space (X, μ) and $q \in [1, \infty]$,

$$g \mapsto \varphi_g : L^q(\mu) \to L^p(\mu)^* \tag{2}$$

is a linear map that preserves norms (for $p = 1$, $q = \infty$ assume (X, μ) is σ-finite). A norm-preserving linear map is an **isometry**; an isometry is automatically injective.

Exercise 6.1.5

For the bounded function $g(x) = x$ on $[0,1]$, show that there is no $f \in L^1[0,1]$, $f \neq 0$, with $\|f\|_1\|g\|_\infty = \int_0^1 fg$.

Solution [Supplement]. Here $\|g\|_\infty = 1$. For any $f \in L^1[0,1]$,

$$\Big|\int_0^1 f(x)x\,dx\Big| \le \int_0^1 |f(x)|x\,dx \le \int_0^1 |f(x)|\,dx = \|f\|_1\|g\|_\infty.$$

Equality would force $\int_0^1 |f(x)|(1-x)\,dx = 0$, so $|f(x)|(1-x) = 0$ a.e.; since $1-x > 0$ on $[0,1)$, $f = 0$ a.e. So the supremum $\|\varphi_g\| = 1$ is not attained. □

6.1.2 The duals of ℓ^p and c_0

In many cases (2) is surjective; we examine ℓ^p. For $n = 1, 2, \ldots$ define the sequence

$$e_n(m) = \begin{cases} 1, & m = n, \\ 0, & m \neq n. \end{cases} \tag{3}$$

If $\varphi \in (\ell^p)^*$ and $\varphi = \varphi_g$, then

$$\varphi(e_n) = \varphi_g(e_n) = \sum_{i=1}^{\infty} g(i)e_n(i) = g(n), \qquad n \in \mathbb{N}.$$

So for arbitrary $\varphi \in (\ell^p)^*$ define $g(n) = \varphi(e_n)$. Assume $1 \le p < \infty$; it suffices to show $g \in \ell^q$ (and $\varphi = \varphi_g$) to prove (2) is surjective.

$p = 1$, $q = \infty$. If $\varphi \in (\ell^1)^*$, then since $e_n \in \ell^1$, $|\varphi(e_n)| \le \|\varphi\|\|e_n\|_1 = \|\varphi\|$ for each n, so $g \in \ell^\infty$ and $\|g\|_\infty \le \|\varphi\|$.

$1 < p < \infty$. Fix $N \in \mathbb{N}$ and, as in Example 2, put

$$f(n) = \begin{cases} |g(n)|^{q-1} \cdot \operatorname{sgn} g(n), & n \le N, \\ 0, & n > N. \end{cases}$$

Then $\|f\|_p^p = \sum_{n=1}^N |f(n)|^p = \sum_{n=1}^N |g(n)|^q$, so

$$\sum_{n=1}^{N} |g(n)|^q = \sum_{n=1}^{\infty} f(n)g(n) = \varphi(f) \le \|\varphi\|\|f\|_p = \|\varphi\|\Big(\sum_{n=1}^{N} |g(n)|^q\Big)^{1/p}.$$

Thus $\big(\sum_{n=1}^N |g(n)|^q\big)^{1/q} \le \|\varphi\|$ for every N, so $\|g\|_q \le \|\varphi\|$ and $g \in \ell^q$.

Finally, to see $\varphi = \varphi_g$, let A be the vector space of finite linear combinations of the e_n. Since $\varphi(e_n) = g(n) = \varphi_g(e_n)$, φ and φ_g agree on A; A is dense in ℓ^p ($p < \infty$) and both are continuous on ℓ^p, so $\varphi_g = \varphi$.

Theorem 6.1.3

If $1 \le p < \infty$ and p, q are conjugate exponents, then $g \mapsto \varphi_g : \ell^q \to (\ell^p)^*$ is a surjective isometry.

Using the Hahn–Banach theorem proved below, one sees that $g \mapsto \varphi_g$ is not surjective for $p = \infty$, i.e. $(\ell^\infty)^* \supsetneq \ell^1$. Since $c_0 = c_0(\mathbb{N})$ is a subspace of ℓ^∞, restricting bounded linear functionals of ℓ^∞ to c_0 gives bounded linear functionals of c_0, so we may consider the linear map

$$g \mapsto \varphi_g : \ell^1 \to (c_0)^*. \tag{4}$$

Exercise 6.1.6

Show that (4) is a surjective isometry.

Solution [Supplement]. *Isometry.* $|\varphi_g(x)| = |\sum g(n)x(n)| \le \|g\|_1\|x\|_\infty$, so $\|\varphi_g\| \le \|g\|_1$. For $N \in \mathbb{N}$ let $x_N(n) = \operatorname{sgn} g(n)$ for $n \le N$ and 0 for $n > N$. Then $x_N \in c_0$, $\|x_N\|_\infty \le 1$, and $\varphi_g(x_N) = \sum_{n\le N}|g(n)|$. Letting $N \to \infty$ gives $\|\varphi_g\| \ge \|g\|_1$. *Surjectivity.* Given $\varphi \in (c_0)^*$, put $g(n) = \varphi(e_n)$. With x_N defined as above for this g, $\sum_{n\le N}|g(n)| = \varphi(x_N) \le \|\varphi\|$, so $g \in \ell^1$ with $\|g\|_1 \le \|\varphi\|$. Now φ and φ_g agree on finite linear combinations of the e_n, which are dense in c_0 (for $x \in c_0$, $\|x - x\chi_{[1,N]}\|_\infty = \sup_{n>N}|x(n)| \to 0$). Both are continuous, so $\varphi = \varphi_g$. □

Summarizing:

$$(c_0)^* = \ell^1, \qquad (\ell^p)^* = \ell^q \quad (1 \le p < \infty),$$

and therefore in many cases $X = X^{**}$.

6.1.3 The canonical embedding $X \to X^{**}$

For each $x \in X$ define

$$T_x(\varphi) = \varphi(x), \qquad \varphi \in X^*. \tag{5}$$

Clearly T_x is a linear functional on X^* and $\|T_x\| \le \|x\|$, so $T_x \in X^{**}$. To show that $x \mapsto T_x$ is injective we must show

$$x \neq 0 \implies T_x \neq 0,$$

i.e. that for $x \neq 0$ there is $\varphi \in X^*$ with $\varphi(x) \neq 0$. Put differently: is there for $x \neq y$ a bounded linear functional $\varphi \in X^*$ with $\varphi(x) \neq \varphi(y)$ — does X have enough bounded linear functionals? The next theorem answers this affirmatively and is fundamental for the relationship between X and X^*.

Theorem 6.1.4 (Hahn–Banach)

Let X be a normed space and Y a subspace. For every $\psi \in Y^*$ there is $\varphi \in X^*$ with

$$\varphi|_Y = \psi, \qquad \|\varphi\| = \|\psi\|.$$

Proof. Although everything so far was over complex vector spaces, it is convenient to prove this first for real vector spaces. So assume X, Y are real and ψ is a real linear functional. If $\psi = 0$ put $\varphi = 0$; so assume $\psi \neq 0$, and after multiplying by a scalar, $\|\psi\| = 1$. The proof has three independent steps. Step 1 is the case where X has dimension one more than Y; it is the heart of the proof. Step 2 uses Step 1 repeatedly, via the axiom of choice. Step 3 extends the real result to complex spaces.

Step 1 (one more dimension). For $x_0 \in X \setminus Y$ consider

$$Y_1 = \{ax_0 + y : a \in \mathbb{R},\ y \in Y\},$$

and ask how to extend ψ to $\psi_1 : Y_1 \to \mathbb{R}$. Since $\psi_1(y) = \psi(y)$ is already fixed, we only need to choose $\psi_1(x_0)$. Any choice gives a linear functional; the point is to get $\|\psi_1\| = 1$, i.e.

$$|a\psi_1(x_0) + \psi(y)| \le \|ax_0 + y\|, \qquad a \in \mathbb{R},\ y \in Y. \tag{6}$$

For $y, y' \in Y$,

$$\psi(y) - \psi(y') = \psi(y - y') \le \|y - y'\| \le \|y - x_0\| + \|y' - x_0\|,$$

so

$$\psi(y) - \|y - x_0\| \le \psi(y') + \|y' - x_0\|, \qquad y, y' \in Y.$$

Hence with

$$A = \sup\{\psi(y) - \|y - x_0\| : y \in Y\}, \qquad B = \inf\{\psi(y') + \|y' - x_0\| : y' \in Y\},$$

we have $A \le B$. Choose any number in $[A, B]$ as $\psi_1(x_0)$. Then for all $y, y' \in Y$

$$\psi(y) - \|y - x_0\| \le \psi_1(x_0) \le \psi(y') + \|y' - x_0\|,$$

which is equivalent to $|\psi_1(x_0) - \psi(y)| \le \|x_0 - y\|$ for all $y \in Y$. This inequality also holds with $-\frac{y}{a} \in Y$ in place of y ($a \neq 0$); multiplying by $|a|$ gives (6) (and (6) is trivial for $a = 0$).

Step 2 (Zorn). Define

$$\mathcal{P} = \{(Y_\iota, \psi_\iota) : \ Y_\iota \text{ a subspace of } X \text{ containing } Y,\ \psi_\iota \in Y_\iota^*,\ \psi_\iota|_Y = \psi,\ \|\psi_\iota\| = 1\}.$$

For $(Y_1, \psi_1), (Y_2, \psi_2) \in \mathcal{P}$ define $(Y_1, \psi_1) \le (Y_2, \psi_2)$ if $Y_1 \subset Y_2$ and $\psi_1 = \psi_2|_{Y_1}$. This makes $\mathcal{P}$ a partially ordered set, so by the axiom of choice (equivalently, the Hausdorff maximality principle) there is a maximal totally ordered subset $\Omega \subset \mathcal{P}$ (maximal meaning: if $\mathcal{N}$ is totally ordered in $\mathcal{P}$ and $\mathcal{N} \supset \Omega$, then $\mathcal{N} = \Omega$). Then

$$Z = \bigcup \{Y_\iota : (Y_\iota, \psi_\iota) \in \Omega\}$$

is a subspace of X, and defining $\rho(x) = \psi_\iota(x)$ when $x \in Y_\iota$ gives a linear functional ρ on Z with $\|\rho\| = 1$ (Exercise 6.1.7). It remains to show $Z = X$. If $Z \subsetneq X$, Step 1 gives $(Z_1, \rho_1) \in \mathcal{P}$ with $Z_1 \supsetneq Z$; then $(Z_1, \rho_1) \notin \Omega$ but $\Omega \cup \{(Z_1, \rho_1)\}$ is totally ordered, contradicting maximality. So $Z = X$, which proves the real case.

Step 3 (complex case). Let X, Y be complex and ψ a complex linear functional. A complex vector space is closed under multiplication by reals, so it may be regarded as a real vector space. Writing $\psi = u + iv$, u is a real linear functional on the real space Y, with $\|u\| \le \|\psi\|$. By the real case there is a real linear functional U on X with $U|_Y = u$ and $\|U\| = \|u\|$. For $x \in X$ define

$$\varphi(x) = U(x) - iU(ix).$$

Clearly $\varphi(x + y) = \varphi(x) + \varphi(y)$, and

$$\varphi(ix) = U(ix) - iU(-x) = i\big[U(x) - iU(ix)\big] = i\varphi(x),$$

so φ is complex linear. For $y \in Y$,

$$\varphi(y) = U(y) - iU(iy) = u(y) - iu(iy),$$

so to show $\varphi|_Y = \psi$ it suffices that $\psi(y) = u(y) - iu(iy)$. For any $\alpha \in \mathbb{C}$ one checks immediately $\alpha = \operatorname{Re}\alpha - i\operatorname{Re}(i\alpha)$; since ψ is complex linear,

$$\psi(y) = \operatorname{Re}\psi(y) - i\operatorname{Re}(i\psi(y)) = u(y) - i\operatorname{Re}(\psi(iy)) = u(y) - iu(iy),$$

so $\varphi|_Y = \psi$.

Finally we show $\|\varphi\| = \|\psi\|$. Since $\|U\| = \|u\| \le \|\psi\|$, it suffices to show $\|\varphi\| \le \|U\|$ (the reverse $\|\varphi\| \ge \|\psi\|$ is clear because φ extends ψ). For $x \in X$ choose $\alpha \in \mathbb{C}$ with $|\alpha| = 1$ and $\alpha\varphi(x) = |\varphi(x)|$. Then $\varphi(\alpha x)$ is real, so

$$|\varphi(x)| = \varphi(\alpha x) = U(\alpha x) \le \|U\|\|\alpha x\| = \|U\|\|x\|.$$

Thus $\|\varphi\| \le \|U\|$, completing the proof. ∎

Exercise 6.1.7

In the proof of the Hahn–Banach theorem, show that Z is a subspace and that ρ is a linear functional on Z with $\|\rho\| = 1$.

Solution [Supplement]. *Z is a subspace.* Let $x, x' \in Z$, say $x \in Y_\iota$, $x' \in Y_\kappa$ with $(Y_\iota, \psi_\iota), (Y_\kappa, \psi_\kappa) \in \Omega$. Since Ω is totally ordered, one contains the other, say $Y_\iota \subset Y_\kappa$; then $ax + bx' \in Y_\kappa \subset Z$. *$\rho$ is well defined.* If $x \in Y_\iota \cap Y_\kappa$ and $Y_\iota \subset Y_\kappa$, then $\psi_\iota = \psi_\kappa|_{Y_\iota}$, so $\psi_\iota(x) = \psi_\kappa(x)$. *Linearity.* With $x, x' \in Y_\kappa$ as above, $\rho(ax + bx') = \psi_\kappa(ax + bx') = a\psi_\kappa(x) + b\psi_\kappa(x') = a\rho(x) + b\rho(x')$. *Norm.* For $x \in Y_\iota$, $|\rho(x)| = |\psi_\iota(x)| \le \|x\|$, so $\|\rho\| \le 1$; and $\rho|_Y = \psi$ with $\|\psi\| = 1$ gives $\|\rho\| \ge 1$. □

For $x \in X$, $x \neq 0$, apply Hahn–Banach to the one-dimensional subspace $Y = \{\alpha x : \alpha \in \mathbb{C}\}$ and the functional $\psi(\alpha x) = \alpha\|x\|$ (which has $\|\psi\| = 1$):

Corollary 6.1.5

If $x \neq 0$ is a point of a normed space X, there is $\varphi \in X^*$ with $\varphi(x) = \|x\|$ and $\|\varphi\| = 1$.

In particular, by Corollary 6.1.5 the norm of T_x defined in (5) equals $\|x\|$, so $x \mapsto T_x$ is an isometry from X into X^{**}. Hence a normed space X may always be regarded as a subspace of X^{**}. If $x \mapsto T_x : X \to X^{**}$ is surjective, X is called **reflexive**. The Banach spaces ℓ^p $(1 < p < \infty)$ are typical reflexive spaces. However $(c_0)^{**} = \ell^\infty$, so c_0 is not reflexive.

The next corollary is useful for deciding whether a given subspace is closed. The closure of a subset Y is written $\overline{Y}$.

Corollary 6.1.6

For a subspace Y of a normed space X and a point $x \in X$, the following are equivalent:

(a) $x \in \overline{Y}$;

(b) if $\varphi \in X^*$ and $\varphi|_Y = 0$, then $\varphi(x) = 0$.

Proof. Every $\varphi \in X^*$ is continuous, so (a)⇒(b) is clear. If $x \notin \overline{Y}$, there is $\delta > 0$ with

$$y \in Y \implies \|x - y\| > \delta.$$

Consider the subspace $Z = \{\alpha x + y : \alpha \in \mathbb{C},\ y \in Y\}$ spanned by x and Y, and define $\psi(\alpha x + y) = \alpha$ (well defined since $x \notin Y$); ψ is a linear functional on Z. For $\alpha x + y \in Z$ with $\alpha \neq 0$,

$$|\psi(\alpha x + y)|\delta = |\alpha|\delta \le |\alpha|\left\|x + \frac{1}{\alpha}y\right\| = \|\alpha x + y\|,$$

so $\psi \in Z^*$ with $\|\psi\| \le \frac{1}{\delta}$. Also $\psi(x) = 1 \neq 0$. Extending ψ to X by Hahn–Banach gives $\varphi \in X^*$ with $\varphi|_Y = 0$ but $\varphi(x) \neq 0$. ■

Exercise 6.1.8

Show that there is $\varphi \in (\ell^\infty)^*$ with $\varphi|_{c_0} = 0$ but $\varphi \neq 0$. Show that $(\ell^\infty)^* \supsetneq \ell^1$, and that the Banach space ℓ^1 is not reflexive.

Solution [Supplement]. c_0 is a closed subspace of ℓ^∞ (Exercise 5.1.9) and the constant sequence $\mathbf{1} \notin c_0$, so $\mathbf{1} \notin \overline{c_0}$. By Corollary 6.1.6 there is $\varphi \in (\ell^\infty)^*$ with $\varphi|_{c_0} = 0$ and $\varphi(\mathbf{1}) \neq 0$. Such φ is not of the form φ_g, $g \in \ell^1$: otherwise $g(n) = \varphi_g(e_n) = \varphi(e_n) = 0$ for all n (as $e_n \in c_0$), so $g = 0$ and $\varphi = 0$. Hence the isometry $g \mapsto \varphi_g : \ell^1 \to (\ell^\infty)^*$ is not surjective: $(\ell^\infty)^* \supsetneq \ell^1$. ℓ^1 not reflexive: $(\ell^1)^* = \ell^\infty$ (Theorem 6.1.3), so $(\ell^1)^{**} = (\ell^\infty)^*$. If $x \mapsto T_x$ were onto, then $\varphi = T_g$ for some $g \in \ell^1$, i.e. $\varphi(h) = T_g(\varphi_h) = \sum_n g(n)h(n)$ for $h \in \ell^\infty$ (identifying h with $\varphi_h \in (\ell^1)^*$); so $\varphi = \varphi_g$, which we just ruled out. □

Exercise 6.1.9

Show that a Banach space X is reflexive if and only if X^* is reflexive. Show that the Banach space ℓ^∞ is not reflexive.

Solution [Supplement]**.** Write $J_X : X \to X^{**}$ and $J_{X^*} : X^* \to X^{***}$ for the canonical isometries.

($\Rightarrow$) Let X be reflexive and $\Phi \in X^{***}$. Put $\varphi = \Phi \circ J_X \in X^*$. For $\xi \in X^{**}$ write $\xi = J_X(x)$; then $\Phi(\xi) = \Phi(J_X x) = \varphi(x) = J_X(x)(\varphi) = \xi(\varphi) = J_{X^*}(\varphi)(\xi)$. So $\Phi = J_{X^*}(\varphi)$ and X^* is reflexive.

($\Leftarrow$) Let X^* be reflexive and suppose $J_X(X) \subsetneq X^{**}$. Since X is complete, $J_X(X)$ is a closed subspace, so by Corollary 6.1.6 there is $\Phi \in X^{***}$ with $\Phi \neq 0$ and $\Phi|_{J_X(X)} = 0$. By reflexivity of X^*, $\Phi = J_{X^*}(\varphi)$ for some $\varphi \in X^*$. Then $\varphi(x) = J_X(x)(\varphi) = \Phi(J_X x) = 0$ for every x, so $\varphi = 0$ and $\Phi = 0$, a contradiction. Finally $\ell^\infty = (\ell^1)^*$ and ℓ^1 is not reflexive (Exercise 6.1.8), so ℓ^∞ is not reflexive. □

6.2 Linear Maps between Banach Spaces

Motivation. Completeness has strong consequences for linear maps, all coming from the Baire category theorem: bijective bounded maps have bounded inverses, closed graphs force boundedness, and pointwise bounded families are uniformly bounded. A striking application is a continuous function whose Fourier series diverges at a point.

So far completeness has played little role. This section shows how completeness matters when the domain or range of a bounded linear map is a Banach space. The following is the key property of complete metric spaces. For a point x of a metric space (X, d) and $r > 0$, write

$$N_X(x, r) = \{y \in X : d(x, y) < r\}$$

for the neighborhood of x, or just $N(x, r)$. When X is a normed space, $N_X(0, r)$ is written X_r.

Theorem 6.2.1 (Baire)

If a sequence $\langle C_n \rangle$ of closed subsets of a complete metric space X has union X, then some C_n contains a nonempty open set.

Proof. Suppose not. Then there is a neighborhood $N(x_1, r_1)$ with $C_1 \cap \overline{N(x_1, r_1)} = \emptyset$ (as $C_1 \neq X$ is closed). Here $\overline{A}$ is the closure of A. By assumption $N(x_1, r_1) \not\subset C_2$, so there is a neighborhood $N(x_2, r_2)$ with

$$\overline{N(x_2, r_2)} \subset N(x_1, r_1) \setminus C_2$$

($N(x_1, r_1) \setminus C_2$ is a nonempty open set). Repeating, we get neighborhoods $\langle N(x_n, r_n) \rangle$ with

$$N(x_{n+1}, r_{n+1}) \subset N(x_n, r_n), \qquad \overline{N(x_n, r_n)} \cap \Big[\bigcup_{k=1}^{n} C_k\Big] = \emptyset$$

for each n. We may also arrange $r_n \to 0$; then $\langle x_n \rangle$ is a Cauchy sequence (for $m > n$, $x_m \in N(x_n, r_n)$). Its limit x lies in every $\overline{N(x_n, r_n)}$, so $x \notin \bigcup_{n=1}^{\infty} C_n = X$, a contradiction. ■

Corollary 6.2.2

In a complete metric space, the intersection of a sequence of dense open sets is nonempty.

Exercise 6.2.1

Prove Corollary 6.2.2.

Solution [Supplement]. Let U_n be dense open sets and $C_n = U_n^c$, closed. Since U_n is dense, C_n contains no nonempty open set. If $\bigcap_n U_n = \emptyset$, then $\bigcup_n C_n = X$, and Baire's theorem gives some C_n containing a nonempty open set — a contradiction. (The same argument applied to the complete space $\overline{N(x,r)}$ shows $\bigcap U_n$ is in fact dense.) □

The core of this section is the open mapping theorem and the uniform boundedness principle; together with Hahn–Banach these are the three basic tools for studying Banach spaces.

6.2.1 The open mapping theorem

Theorem 6.2.3 (open mapping theorem)

Let $T : X \to Y$ be a surjective bounded linear map between Banach spaces. Then there is $\delta > 0$ with $T(X_1) \supset Y_\delta$.

Proof. First, $X = \bigcup_n X_n$ and T is surjective, so $Y = \bigcup_n \overline{T(X_n)}$. By Baire's theorem there are $n \in \mathbb{N}$ and a neighborhood $N_Y(y_0, r)$ with $N_Y(y_0, r) \subset \overline{T(X_n)}$. If $\|y\| < r$, then $y + y_0 \in N_Y(y_0, r)$, so $y + y_0 \in \overline{T(X_n)}$. Choosing x_0 with $T(x_0) = y_0$,

$$y \in \overline{T(X_n)} - T(x_0) \subset \overline{T(X_{n+\|x_0\|})}.$$

Putting $\delta = \dfrac{r}{n + \|x_0\|}$ we get $Y_\delta \subset \overline{T(X_1)}$, and by homogeneity

$$Y_s \subset \overline{T(X_{s/\delta})}, \qquad s \in (0, \infty). \tag{7}$$

We now use (7) to show $Y_\delta \subset T(X_1)$ up to a factor. Fix $\epsilon \in (0,1)$. If $y \in Y_1$, by (7) there is $x_1 \in X_{1/\delta}$ with $\|y - T(x_1)\| < \epsilon$. Since $y - T(x_1) \in Y_\epsilon$, (7) with $s = \epsilon$ gives $x_2 \in X_{\epsilon/\delta}$ with $\|y - T(x_1) - T(x_2)\| < \epsilon^2$. Continuing, we obtain $\langle x_n \rangle$ with

$$\|x_n\| < \frac{\epsilon^{n-1}}{\delta}, \qquad \|y - T(x_1) - T(x_2) - \cdots - T(x_n)\| < \epsilon^n.$$

Hence $\sum_{n=1}^{\infty} x_n$ converges (absolutely, in the Banach space X) to some $x \in X$ with $\|x\| \le \frac{1}{\delta(1-\epsilon)}$, and

$$y = \lim_{n\to\infty} T(x_1 + x_2 + \cdots + x_n) = T(x).$$

So $y \in T(X_{1/\delta(1-\epsilon)})$; thus $Y_1 \subset T(X_{1/\delta(1-\epsilon)})$, i.e. $Y_{\delta(1-\epsilon)} \subset T(X_1)$. ■

A map $f : X \to Y$ between metric spaces is an **open map** if $f(U)$ is open in Y for every open $U \subset X$. The next corollary explains the name.

Corollary 6.2.4

A surjective bounded linear map $T : X \to Y$ between Banach spaces is always an open map.

Exercise 6.2.2

Prove Corollary 6.2.4.

Solution [Supplement]. Let $U \subset X$ be open and $y = T(x) \in T(U)$ with $x \in U$. Choose $r > 0$ with $N_X(x, r) = x + X_r \subset U$. By Theorem 6.2.3, $T(X_r) = rT(X_1) \supset rY_\delta = Y_{r\delta}$. Hence

$$N_Y(y, r\delta) = y + Y_{r\delta} \subset T(x) + T(X_r) = T(x + X_r) \subset T(U),$$

so $T(U)$ is open. □

Corollary 6.2.5

If a bounded linear map $T : X \to Y$ between Banach spaces is bijective, then its inverse $T^{-1} : Y \to X$ is also bounded.

Proof. Choose $\delta > 0$ with $Y_\delta \subset T(X_1)$. Then $T^{-1}(Y_\delta) \subset X_1$, so $\|T^{-1}\| \le \frac{1}{\delta}$. ■

6.2.2 The Fourier transform $L^1(\mathbb{T}) \to c_0(\mathbb{Z})$ is not onto

For $n = 1, 2, \dots$ let

$$D_n(t) = \sum_{k=-n}^{n} u_k(t) = \frac{\sin(n + \frac{1}{2})t}{\sin \frac{1}{2}t}, \qquad t \in \mathbb{T}. \tag{8}$$

$\langle D_n \rangle$ is the **Dirichlet kernel**.

Exercise 6.2.3

Prove the second equality in (8).

Solution [Supplement]**.** For $t \neq 0 \pmod{2\pi}$, the geometric sum gives

$$\sum_{k=-n}^{n} e^{ikt} = e^{-int}\frac{e^{i(2n+1)t} - 1}{e^{it} - 1} = \frac{e^{i(n+1)t} - e^{-int}}{e^{it} - 1} = \frac{e^{i(n+\frac{1}{2})t} - e^{-i(n+\frac{1}{2})t}}{e^{it/2} - e^{-it/2}} = \frac{\sin(n + \frac{1}{2})t}{\sin \frac{1}{2}t},$$

after multiplying numerator and denominator by $e^{-it/2}$. At $t = 0$ both sides equal $2n + 1$ (as a limit). □

Then $\widehat{D_n} = \sum_{k=-n}^{n} \chi_{\{k\}}$, so $\|\widehat{D_n}\|_\infty = 1$. On the other hand, since $|\sin t| \le |t|$,

$$\begin{aligned}\|D_n\|_1 &\ge \frac{1}{2\pi}\int_{-\pi}^{\pi} \Big|\frac{\sin(n + \frac{1}{2})t}{\frac{t}{2}}\Big| dt = \frac{2}{\pi}\int_0^{\pi} \frac{1}{t}\Big|\sin\Big(n + \frac{1}{2}\Big)t\Big| dt = \frac{2}{\pi}\int_0^{(n+\frac{1}{2})\pi} \frac{1}{t}|\sin t|\, dt \\ &\ge \frac{2}{\pi}\sum_{k=1}^{n} \frac{1}{k\pi}\int_{(k-1)\pi}^{k\pi} |\sin t|\, dt = \frac{4}{\pi^2}\sum_{k=1}^{n}\frac{1}{k},\end{aligned}$$

so $\lim_{n\to\infty}\|D_n\|_1 = \infty$. With $E_n = D_n/\|D_n\|_1$ we have $\|E_n\|_1 = 1$ but $\|\widehat{E_n}\|_\infty \to 0$. So the Fourier transform $f \mapsto \widehat{f}$ cannot satisfy the conclusion of the open mapping theorem (no $\delta > 0$ with $\|\widehat{f}\|_\infty \ge \delta\|f\|_1$, which Corollary 6.2.5 would give since $f \mapsto \widehat{f}$ is injective). Hence *the Fourier transform $L^1(\mathbb{T}) \to c_0(\mathbb{Z})$ is not surjective.*

6.2.3 The closed graph theorem

Another application of the open mapping theorem is the closed graph theorem, which is essential for proving that a given linear map is bounded. The hypothesis says that the graph of T is closed.

Theorem 6.2.6 (closed graph theorem)

A linear map $T : X \to Y$ between Banach spaces satisfying

$$x_n \in X,\ \lim_{n\to\infty} x_n = x,\ \lim_{n\to\infty} T(x_n) = y \implies T(x) = y$$

is bounded.

Proof. For $x \in X$ put $\|x\|' = \|x\| + \|T(x)\|$; then $x \mapsto \|x\|'$ is a norm on X. If $\langle x_n \rangle$ is Cauchy for $(X, \|\cdot\|')$, then $\langle x_n \rangle$ is Cauchy in the Banach space $(X, \|\cdot\|)$ and $\langle T(x_n) \rangle$ is Cauchy in the Banach space Y. So there are $x \in X$, $y \in Y$ with

$$\lim_{n\to\infty} \|x_n - x\| = 0, \qquad \lim_{n\to\infty} \|T(x_n) - y\| = 0.$$

By hypothesis $T(x) = y$, so

$$\lim_{n\to\infty} \|x_n - x\|' = \lim_{n\to\infty} \|x_n - x\| + \lim_{n\to\infty} \|T(x_n) - T(x)\| = 0,$$

and $(X, \|\cdot\|')$ is a Banach space. Since $\|x\| \le \|x\|'$, the identity $\iota : (X, \|\cdot\|') \to (X, \|\cdot\|)$ is a bounded bijection, so by Corollary 6.2.5 its inverse $\iota^{-1} : (X, \|\cdot\|) \to (X, \|\cdot\|')$ is bounded. Hence there is $M > 0$ with $\|x\|' \le M\|x\|$, and

$$\|Tx\| + \|x\| = \|x\|' \le M\|x\|, \qquad x \in X,$$

so $\|Tx\| \le (M-1)\|x\|$. ∎

Exercise 6.2.4

In the proof of the closed graph theorem, show that $x \mapsto \|x\|'$ is a norm.

Solution [Supplement]. $\|x\|' \ge 0$, and $\|x\|' = 0$ forces $\|x\| = 0$, i.e. $x = 0$. By linearity of T, $\|ax\|' = |a|\|x\| + |a|\|Tx\| = |a|\|x\|'$ and

$$\|x + z\|' = \|x + z\| + \|Tx + Tz\| \le \|x\| + \|z\| + \|Tx\| + \|Tz\| = \|x\|' + \|z\|'.$$

□

6.2.4 The uniform boundedness principle

Another application of Baire's theorem is the uniform boundedness principle, also called the Banach–Steinhaus theorem.

Theorem 6.2.7 (uniform boundedness principle)

Let $\{T_\alpha : \alpha \in A\}$ be a family of bounded linear maps from a Banach space X to a normed space Y. If for every $x \in X$

$$M(x) = \sup\{\|T_\alpha(x)\| : \alpha \in A\} < \infty,$$

then $\sup\{\|T_\alpha\| : \alpha \in A\} < \infty$.

Proof. For $n = 1, 2, \dots$ put

$$C_n = \bigcap_{\alpha \in A} \{x \in X : \|T_\alpha(x)\| \le n\}.$$

Each $x \mapsto \|T_\alpha(x)\|$ is continuous, so C_n is closed, and by hypothesis $X = \bigcup_n C_n$. By Baire's theorem some C_n contains a neighborhood $N(x_0, r)$, i.e.

$$\alpha \in A, \ \|x - x_0\| < r \implies \|T_\alpha(x)\| \le n.$$

If $\|x\| < r$, then $\|(x + x_0) - x_0\| < r$, so for every $\alpha \in A$

$$\|T_\alpha(x)\| - \|T_\alpha(x_0)\| \le \|T_\alpha(x + x_0)\| \le n,$$

hence $\|T_\alpha(x)\| \le n + \|T_\alpha(x_0)\| \le n + M(x_0)$. Therefore

$$\sup_{\alpha\in A}\|T_\alpha\| \le \frac{1}{r}(n + M(x_0)).$$

■

The next corollary is very useful for defining new bounded linear maps from a given sequence.

Corollary 6.2.8

Let $\langle T_n\rangle$ be a sequence of bounded linear maps from a Banach space X to a normed space Y such that $\langle T_n(x)\rangle$ converges in Y for every $x \in X$. Then $T(x) = \lim_{n\to\infty} T_n(x)$ defines a bounded linear map $T : X \to Y$.

Proof. T is clearly linear, and $\{T_n\}$ satisfies the hypothesis of the uniform boundedness principle (convergent sequences are bounded). So with $M = \sup_n\|T_n\| < \infty$, for each $x \in X$

$$\|T(x)\| = \lim_{n\to\infty}\|T_n(x)\| \le \sup_n\|T_n\|\|x\| = M\|x\|,$$

so $\|T\| \le M$. ■

6.2.5 A continuous function whose Fourier series diverges

Using the uniform boundedness principle we show that the Fourier series of a continuous periodic function need not converge pointwise. By (2) of §5.1 the partial sums of the Fourier series are

$$\sum_{k=-n}^{n}\widehat{f}(k)u_k = \sum_{k=-n}^{n} u_k * f = D_n * f.$$

Fix $t_0 \in \mathbb{T}$ and n, and consider the linear functional on the Banach space $C(\mathbb{T})$

$$T_n : f \mapsto (D_n * f)(t_0), \qquad f \in C(\mathbb{T}).$$

Then

$$|T_n(f)| \le \int_{\mathbb{T}}|f(t)D_n(t_0 - t)|\,dt \le \|f\|_\infty\|D_n\|_1, \qquad f \in C(\mathbb{T}),$$

so $\|T_n\| \le \|D_n\|_1$; in fact $\|T_n\| = \|D_n\|_1$. The function $t \mapsto \operatorname{sgn} D_n(t_0 - t)$ has only finitely many discontinuities, so we can find continuous functions $\langle h_i\rangle$ with $\|h_i\|_\infty \le 1$ converging pointwise to it (except at those finitely many points). By the Lebesgue convergence theorem

$$T_n(h_i) = \int_{\mathbb{T}} h_i(t)D_n(t_0 - t)\,dt \to \int_{\mathbb{T}}|D_n(t_0 - t)|\,dt = \|D_n\|_1,$$

so $\|T_n\| = \|D_n\|_1$. If the Fourier series of every continuous function converged at t_0, then $\langle T_n(f)\rangle$ would converge to $f(t_0)$ for each $f \in C(\mathbb{T})$, and the uniform boundedness principle would make $\langle\|T_n\|\rangle$ bounded — but we know $\|D_n\|_1 \to \infty$. Hence *there is a continuous function whose Fourier series does not converge pointwise.*

6.3 Hilbert Spaces

Motivation. Inner products bring Euclidean geometry (angles, orthogonal projections, Pythagoras) to infinite dimensions. In a Hilbert space every functional is an inner product, every vector expands in an orthonormal basis, and for $L^2(\mathbb{T})$ this is exactly the theory of Fourier series.

6.3.1 Inner products

Definition (inner product)

For a complex vector space X, a function $(x,y)\mapsto\langle x,y\rangle : X\times X\to\mathbb{C}$ satisfying

(I1) $x\mapsto\langle x,y\rangle : X\to\mathbb{C}$ is a linear functional for each $y\in X$;

(I2) $\langle x,y\rangle=\overline{\langle y,x\rangle}$ for all $x,y\in X$;

(I3) $\langle x,x\rangle\geq 0$ for all $x\in X$;

(I4) $x=0 \iff \langle x,x\rangle=0$

is an **inner product**, and a vector space with an inner product is an **inner product space**.

Example 1

On $L^2(X,\mu)$,

$$\langle f,g\rangle=\int_X f\overline{g}\,d\mu,\qquad f,g\in L^2(X,\mu),$$

is an inner product.

For x,y in an inner product space X, choose $\alpha\in\mathbb{C}$ with $|\alpha|=1$ and $\alpha\langle y,x\rangle=|\langle x,y\rangle|$. Then for every real t,

$$0\leq\langle x-t\alpha y,x-t\alpha y\rangle=\langle x,x\rangle-t\alpha\langle y,x\rangle-t\overline{\alpha}\langle x,y\rangle+t^2\langle y,y\rangle=\langle x,x\rangle-2t|\langle x,y\rangle|+t^2\langle y,y\rangle.$$

A nonnegative real quadratic in t has nonpositive discriminant, so

$$|\langle x,y\rangle|^2\leq\langle x,x\rangle\langle y,y\rangle,\qquad x,y\in X.$$

This is the **Schwarz inequality**; for $L^2(X,\mu)$ it is a special case of Hölder's inequality. For $x\in X$ put

$$\|x\|=\langle x,x\rangle^{1/2};\tag{9}$$

$x\mapsto\|x\|$ is easily seen to be a norm. For instance the triangle inequality $\|x+y\|\leq\|x\|+\|y\|$ follows from Schwarz: $\|x+y\|^2=\|x\|^2+2\,\mathrm{Re}\langle x,y\rangle+\|y\|^2\leq(\|x\|+\|y\|)^2$. Hence every inner product space is a normed space. A complete inner product space is a **Hilbert space**. Conversely the norm determines the inner product via the **polarization identity**

$$\langle x,y\rangle=\frac{1}{4}\sum_{k=0}^{3}i^k\|x+i^ky\|^2,\tag{10}$$

and the norm of an inner product space satisfies the **parallelogram law**

$$\|x+y\|^2+\|x-y\|^2=2\|x\|^2+2\|y\|^2.\tag{11}$$

Exercise 6.3.1

Prove (10) and (11) for vectors x,y of an inner product space X.

Solution [Supplement]**.** For $c\in\mathbb{C}$ with $|c|=1$,

$$\|x+cy\|^2=\|x\|^2+\overline{c}\langle x,y\rangle+c\langle y,x\rangle+\|y\|^2.$$

(11): add the cases $c=1$ and $c=-1$; the cross terms cancel. (10): multiply the case $c=i^k$ by i^k and sum over $k=0,1,2,3$:

$$\sum_k i^k\|x+i^ky\|^2=\Big(\sum_k i^k\Big)(\|x\|^2+\|y\|^2)+\Big(\sum_k i^k\overline{i^k}\Big)\langle x,y\rangle+\Big(\sum_k i^{2k}\Big)\langle y,x\rangle=0+4\langle x,y\rangle+0,$$

since $\sum_{k=0}^{3}i^k=0$, $i^k\overline{i^k}=1$, and $\sum_k(-1)^k=0$. □

6.3.2 The dual of a Hilbert space

We now determine the dual of a Hilbert space H. By the Schwarz inequality, for each $y \in H$

$$\phi_y : x \mapsto \langle x, y\rangle, \qquad x \in H,$$

is a bounded linear functional with $\|\phi_y\| = \|y\|$ (take $x = y$ for equality). Hence

$$y \mapsto \phi_y : H \to H^* \tag{12}$$

is an isometric injection. Note that this map is not linear: it preserves sums, but $\phi_{\alpha y} = \overline{\alpha}\phi_y$ for $\alpha \in \mathbb{C}$. Such a map is called **conjugate linear**.

We want to show (12) is surjective: for every $\phi \in H^*$ find $y \in H$ with $\phi(x) = \langle x, y\rangle$. Intuitively, it suffices to find a vector perpendicular to the subspace $\ker\phi$; the next theorem guarantees such vectors exist.

Theorem 6.3.1 (closest point)

Let C be a closed convex subset of a Hilbert space H and $x_0 \in H \setminus C$. Put

$$d = \inf\{\|x_0 - y\| : y \in C\}.$$

Then there is a unique $y_0 \in C$ with $\|x_0 - y_0\| = d$.

Proof. We may assume $x_0 = 0$ (translate C by $-x_0$). Replacing x, y by $\frac{1}{2}x, \frac{1}{2}y$ in the parallelogram law gives

$$\frac{1}{4}\|x - y\|^2 = \frac{1}{2}\|x\|^2 + \frac{1}{2}\|y\|^2 - \left\|\frac{x+y}{2}\right\|^2.$$

If $x, y \in C$ then $\frac{x+y}{2} \in C$ by convexity, so $\|\frac{x+y}{2}\| \geq d$ and

$$\|x - y\|^2 \leq 2\|x\|^2 + 2\|y\|^2 - 4d^2, \qquad x, y \in C. \tag{13}$$

Uniqueness. If $x, y \in C$ and $\|x\| = \|y\| = d$, (13) gives $x = y$. *Existence.* Take a sequence $\langle y_n\rangle$ in C with $\|y_n\| \to d$. By (13),

$$\|y_n - y_m\|^2 \leq 2\|y_n\|^2 + 2\|y_m\|^2 - 4d^2 \to 0,$$

so $\langle y_n\rangle$ is Cauchy. As H is complete and C closed, $y_0 = \lim_n y_n \in C$, and $\|y_0\| = \lim_n \|y_n\| = d$. ■

Now let E be a closed subspace of H and $x_0 \in H \setminus E$. By Theorem 6.3.1 there is $y_0 \in E$ with

$$y \in E \implies \|x_0 - y_0\| \leq \|x_0 - y\|.$$

Put $z_0 = x_0 - y_0$. For every $y \in E$ and $\alpha \in \mathbb{C}$,

$$\langle z_0, z_0\rangle = \|x_0 - y_0\|^2 \leq \|x_0 - y_0 - \alpha y\|^2 = \langle z_0 - \alpha y, z_0 - \alpha y\rangle,$$

so

$$0 \leq -\alpha\langle y, z_0\rangle - \overline{\alpha}\langle z_0, y\rangle + |\alpha|^2\langle y, y\rangle.$$

Substituting $\alpha = t\langle z_0, y\rangle$ $(t \in \mathbb{R})$,

$$0 \leq -2t|\langle y, z_0\rangle|^2 + t^2|\langle y, z_0\rangle|^2\langle y, y\rangle$$

for every $t \in \mathbb{R}$; taking small $t > 0$ forces $\langle y, z_0\rangle = 0$ for every $y \in E$. Two vectors with $\langle x, y\rangle = 0$ are **orthogonal**, written $x \perp y$; for a subset E, $E^\perp$ denotes the set of vectors orthogonal to every vector of E. Since $z_0 \neq 0$ (as $x_0 \notin E$) and $z_0 \in E^\perp$:

Corollary 6.3.2

If E is a closed proper subspace of a Hilbert space H, then $E^\perp \supsetneq \{0\}$.

Return to $\phi \in H^*$. If $\phi \neq 0$, apply Corollary 6.3.2 to the closed proper subspace $\ker\phi$ to get $z \in (\ker\phi)^\perp$ with $\|z\| = 1$. Since $\phi(x)z - \phi(z)x \in \ker\phi$, for every $x \in H$

$$0 = \langle \phi(x)z - \phi(z)x, z\rangle = \phi(x) - \phi(z)\langle x, z\rangle = \phi(x) - \langle x, \overline{\phi(z)}z\rangle.$$

So with $y = \overline{\phi(z)}z$,

$$\phi(x) = \langle x, y\rangle, \qquad x \in H.$$

This proves:

Theorem 6.3.3 (Riesz representation)

For a Hilbert space H, the map $y \mapsto \phi_y$ defined by (12) is a surjective, isometric, conjugate linear map $H \to H^*$.

Exercise 6.3.2

Show that every Hilbert space is reflexive.

Solution [Supplement]**.** By Theorem 6.3.3, H^* is itself a Hilbert space with inner product $\langle \phi_y, \phi_z\rangle_{H^*} = \langle z, y\rangle$ (this is an inner product inducing the dual norm, since $y \mapsto \phi_y$ is a conjugate-linear isometric bijection). Let $\Lambda \in H^{**}$. By Theorem 6.3.3 applied to H^*, there is $\phi_z \in H^*$ with $\Lambda(\phi) = \langle \phi, \phi_z\rangle_{H^*}$ for all $\phi \in H^*$. Then for every y,

$$\Lambda(\phi_y) = \langle \phi_y, \phi_z\rangle_{H^*} = \langle z, y\rangle = \phi_y(z) = T_z(\phi_y).$$

Since every element of H^* is some ϕ_y, $\Lambda = T_z$. So $x \mapsto T_x$ is onto. □

Exercise 6.3.3

Show that $E^\perp$ is a closed subspace for any subset E of a Hilbert space. Show that $\overline{E} = E^{\perp\perp}$ for any subspace E.

Solution [Supplement]**.** $E^\perp = \bigcap_{y\in E} \ker\phi_y$ is an intersection of kernels of bounded linear functionals, hence a closed subspace. For a subspace E: every $x \in E$ is orthogonal to $E^\perp$, so $E \subset E^{\perp\perp}$, and since $E^{\perp\perp}$ is closed, $\overline{E} \subset E^{\perp\perp}$. Conversely let $x \in E^{\perp\perp}$. $\overline{E}$ is a closed subspace; if $x \notin \overline{E}$, the closest-point argument gives $y_0 \in \overline{E}$ with $z_0 = x - y_0 \neq 0$ and $z_0 \perp \overline{E}$, so $z_0 \in E^\perp$. Since $x \in E^{\perp\perp}$ and $y_0 \in \overline{E} \subset E^{\perp\perp}$, also $z_0 \in E^{\perp\perp}$. So $z_0 \perp z_0$, i.e. $z_0 = 0$ — a contradiction. Hence $E^{\perp\perp} = \overline{E}$. □

6.3.3 Orthonormal sets

A subset $\{u_\alpha : \alpha \in A\}$ of an inner product space H satisfying

$$\langle u_\alpha, u_\beta\rangle = \begin{cases} 1, & \alpha = \beta, \\ 0, & \alpha \neq \beta \end{cases}$$

is an **orthonormal set**; the subspace it spans (finite linear combinations) is written P_A.

Proposition 6.3.4

Let $\{u_\alpha : \alpha \in F\}$ be a finite subset of an orthonormal set $\{u_\alpha : \alpha \in A\}$ in an inner product space H. For fixed $x \in H$,

$$\Big\|x - \sum_{\alpha\in F} \langle x, u_\alpha\rangle u_\alpha\Big\| \le \|x - y\|, \qquad y \in P_F, \tag{14}$$

with equality only for $y = \sum_{\alpha\in F}\langle x, u_\alpha\rangle u_\alpha$. Moreover

$$\sum_{\alpha\in F} |\langle x, u_\alpha\rangle|^2 \le \|x\|^2, \qquad x \in H.$$

Proof. Write $z = \sum_{\alpha\in F}\langle x, u_\alpha\rangle u_\alpha$. For each $\alpha \in F$, $\langle x - z, u_\alpha\rangle = \langle x, u_\alpha\rangle - \langle x, u_\alpha\rangle = 0$. If $y \in P_F$ then $y - z \in P_F$, so $\langle x - z, y - z\rangle = 0$, and

$$\|x - y\|^2 = \|x - z\|^2 + \|z - y\|^2, \qquad y \in P_F,$$

which proves the first part (with equality iff $y = z$). Also $\|z\|^2 = \sum_{\alpha\in F}|\langle x, u_\alpha\rangle|^2$, so putting $y = 0$ gives $\|x\|^2 = \|x - z\|^2 + \|z\|^2 \ge \|z\|^2$, the second inequality. ∎

Inequality (14) says that the coefficients c_α making $\sum_\alpha c_\alpha u_\alpha$ closest to x are $c_\alpha = \langle x, u_\alpha\rangle$. Since the second inequality holds for every finite $F \subset A$, Fact B.28 gives:

Corollary 6.3.5 (Bessel's inequality)

For any orthonormal set $\{u_\alpha : \alpha \in A\}$ in an inner product space H,

$$\sum_{\alpha\in A} |\langle x, u_\alpha\rangle|^2 \le \|x\|^2, \qquad x \in H.$$

We want to know for which orthonormal sets Bessel's inequality is an equality. In finite dimensions the orthonormal set must be a basis; we explain the analogue for general Hilbert spaces.

Theorem 6.3.6

For an orthonormal set $\mathcal{U} = \{u_\alpha : \alpha \in A\}$ in a Hilbert space H the following are equivalent:

(a) for every $x \in H$,
$$\sum_{\alpha\in A} |\langle x, u_\alpha\rangle|^2 = \|x\|^2; \tag{15}$$

(b) for all $x, y \in H$,
$$\sum_{\alpha\in A} \langle x, u_\alpha\rangle\overline{\langle y, u_\alpha\rangle} = \langle x, y\rangle; \tag{16}$$

(c) if $\mathcal{W} \supset \mathcal{U}$ is an orthonormal set, then $\mathcal{W} = \mathcal{U}$ (maximality);

(d) P_A is dense in H.

Proof. For each $x \in H$ define $\widehat{x} : A \to \mathbb{C}$ by

$$\widehat{x}(\alpha) = \langle x, u_\alpha\rangle, \qquad \alpha \in A.$$

By Bessel's inequality $\widehat{x} \in \ell^2(A)$, and we get a bounded linear map

$$\Phi : x \mapsto \widehat{x} : H \to \ell^2(A). \tag{17}$$

(a) says Φ preserves norms; such a map preserves inner products by the polarization identity (10). The left side of (16) is exactly $\langle \widehat{x}, \widehat{y}\rangle$, so (a)⇒(b).

(b)⇒(c): if $\mathcal{W} \supsetneq \mathcal{U}$ is orthonormal, pick $u \in \mathcal{W} \setminus \mathcal{U}$; then (16) with $x = y = u$ reads $0 = \sum_\alpha |\langle u, u_\alpha\rangle|^2 = \|u\|^2 = 1$, a contradiction.

(c)⇒(d): if P_A is not dense in H, Corollary 6.3.2 gives a unit vector $u \in (P_A)^\perp = (\overline{P_A})^\perp$, and $\mathcal{U} \cup \{u\}$ is a larger orthonormal set, so (c) fails.

(d)⇒(a): For $x \in P_A$ (a finite linear combination of the u_α) (15) holds, so Φ is an isometry on the dense subspace P_A. For arbitrary $x \in H$ choose $x_n \in P_A$ with $x_n \to x$. Since Φ is bounded,

$$\sum_{\alpha\in A} |\langle x, u_\alpha\rangle|^2 = \|\widehat{x}\|_2^2 = \lim_{n\to\infty} \|\widehat{x_n}\|_2^2 = \lim_{n\to\infty} \|x_n\|^2 = \|x\|^2.$$

■

Identity (16) is **Parseval's identity**. An orthonormal set $\{u_\alpha : \alpha \in A\}$ satisfying the conditions of Theorem 6.3.6 is an **orthonormal basis** of H. Given an orthonormal basis, the linear map Φ of (17) preserves inner products. A linear map between Hilbert spaces preserving inner products is a **Hilbert space isomorphism**. Let us show Φ is surjective. Given $y \in \ell^2(A)$, choose finitely supported $y_n \in \ell^2(A)$ with $\|y_n - y\|_2 \to 0$. Then $\Phi(x_n) = y_n$ for some $x_n \in P_A$, and $\|x_n - x_m\| = \|y_n - y_m\|_2$, so $\langle x_n\rangle$ is Cauchy in H. If $x \in H$ is its limit, $\Phi(x) = \lim_n \Phi(x_n) = \lim_n y_n = y$. Hence:

Theorem 6.3.7

If $\{u_\alpha : \alpha \in A\}$ is an orthonormal basis of a Hilbert space H, then $x \mapsto \widehat{x}$ is a bijective Hilbert space isomorphism $H \to \ell^2(A)$.

Thus the structure of a Hilbert space is completely determined by the cardinality of an orthonormal basis. Of course, which orthonormal set is convenient depends on the situation.

6.3.4 Fourier series in $L^2(\mathbb{T})$

The set $\{u_n : n \in \mathbb{Z}\}$ of §5.1 (1) is clearly an orthonormal set in $L^2(\mathbb{T})$, and $\widehat{f}(n) = \langle f, u_n\rangle$ for $f \in L^2(\mathbb{T})$. Corollary 5.1.13 says that Theorem 6.3.6(d) holds in this case, so $\{u_n : n \in \mathbb{Z}\}$ is an orthonormal basis of $L^2(\mathbb{T})$. Writing out the identities:

$$\sum_{n=-\infty}^{\infty} \widehat{f}(n)\overline{\widehat{g}(n)} = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)\overline{g(t)}\,dt, \qquad f, g \in L^2(\mathbb{T}),$$

also called **Parseval's identity** (the series converges absolutely for $f, g \in L^2(\mathbb{T})$, so the limit is unambiguous). Moreover

$$\lim_{n\to\infty} \Big\| f - \sum_{k=-n}^{n} \widehat{f}(k)u_k \Big\|_2 = 0, \qquad f \in L^2(\mathbb{T}). \tag{18}$$

(The partial sums of the Fourier series converge in $\|\cdot\|_2$, so by Fact B.18 some subsequence converges almost everywhere. In fact, in the 1960s it was proved that the Fourier series of an L^2 periodic function converges almost everywhere (Carleson). On the other hand, there is an L^1 periodic function whose Fourier series diverges everywhere (Kolmogorov).)

In Theorem 6.3.6 completeness of the Hilbert space played no big role, as the proof shows. But applying Theorem 6.3.7 to Fourier series says that for every $a \in \ell^2(\mathbb{Z})$ there is an L^2-periodic function with Fourier coefficients a; here the completeness of $L^2(\mathbb{T})$ is decisive. Together with Fact B.17 this is called the **Riesz–Fischer theorem**. If $\int_{\mathbb{T}} |f|^2 < \infty$ were understood as an improper Riemann integral, this conclusion would be unobtainable — one of the big reasons the Lebesgue integral is more useful than the Riemann integral.

6.3.5 The Plancherel transform

By Fact B.29, $L^1(\mathbb{T}) \supset L^2(\mathbb{T})$, so restricting the Fourier transform to $L^2(\mathbb{T})$ gives the Hilbert space isomorphism $f \mapsto \widehat{f}$ and Parseval's identity (18). However $L^1(\mathbb{R})$ no longer contains $L^2(\mathbb{R})$, so to get the corresponding theorem we start from $C_c(\mathbb{R}) \subset L^1(\mathbb{R}) \cap L^2(\mathbb{R})$.

If $f \in C_c(\mathbb{R})$ and $\operatorname{supp} f \subset (-\pi, \pi)$, then Parseval for periodic functions and (19) of §5.2 (with $F = f$ on $(-\pi, \pi)$, $\widehat{F}(n) = \frac{1}{2\pi}\widehat{f}(n)$) give

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} |f(t)|^2\, dt = \sum_{n=-\infty}^{\infty} \Big|\frac{1}{2\pi}\widehat{f}(n)\Big|^2.$$

This also holds for $g(x) = f(x)e^{-i\alpha x}$ (with $\widehat{g}(n) = \widehat{f}(n+\alpha)$), so

$$\int_{\mathbb{R}} |f(x)|^2\, dx = \frac{1}{2\pi}\sum_{n=-\infty}^{\infty} |\widehat{f}(n+\alpha)|^2.$$

Integrating both sides over $\alpha \in [0, 1)$,

$$\int_{\mathbb{R}} |f(x)|^2\, dx = \frac{1}{2\pi}\int_0^1 \sum_{n=-\infty}^{\infty} |\widehat{f}(n+\alpha)|^2\, d\alpha = \frac{1}{2\pi}\int_{\mathbb{R}} |\widehat{f}(\alpha)|^2\, d\alpha.$$

In general, if $\operatorname{supp} f \subset (-\pi\lambda, \pi\lambda)$, put $g(x) = \sqrt{\lambda} f(\lambda x)$; then $\operatorname{supp} g \subset (-\pi, \pi)$ and $\widehat{g}(\alpha) = \frac{1}{\sqrt{\lambda}}\widehat{f}(\alpha/\lambda)$, so for every $f \in C_c(\mathbb{R})$

$$\int_{\mathbb{R}} |f(x)|^2\, dx = \int_{\mathbb{R}} |g(x)|^2 = \frac{1}{2\pi}\int_{\mathbb{R}} |\widehat{g}(\alpha)|^2\, d\alpha = \frac{1}{2\pi}\int_{\mathbb{R}} |\widehat{f}(\alpha)|^2\, d\alpha.$$

Let $\widehat{\mathbb{R}}$ denote the real line with $\frac{1}{2\pi}$ times Lebesgue measure. Then

$$\int_{\mathbb{R}} |f(x)|^2\, dx = \int_{\widehat{\mathbb{R}}} |\widehat{f}(\alpha)|^2\, d\alpha, \qquad f \in C_c(\mathbb{R}), \tag{19}$$

and the inversion formula reads

$$f(x) = \int_{\widehat{\mathbb{R}}} \widehat{f}(\alpha) e^{i\alpha x}\, d\alpha.$$

Regarding $C_c(\mathbb{R})$ as a subspace of $L^2(\mathbb{R})$, (19) gives an isometric linear map

$$f \mapsto \widehat{f} : C_c(\mathbb{R}) \to L^2(\widehat{\mathbb{R}}).$$

Since $C_c(\mathbb{R})$ is dense in $L^2(\mathbb{R})$, this map extends to an isometric linear map

$$\mathcal{F} : L^2(\mathbb{R}) \to L^2(\widehat{\mathbb{R}}).$$

Consistency on $L^1 \cap L^2$. If $f \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, both $\widehat{f}$ and $\mathcal{F}(f)$ are defined; is $\widehat{f} = \mathcal{F}(f)$? Review the proofs of Fact B.20, Fact B.21 and Fact B.22. If $p \neq q \in [1, \infty)$ and $f \in L^p[a, b] \cap L^q[a, b]$, there is a simple function $s : [a, b] \to \mathbb{C}$ with simultaneously

$$\|f - s\|_p < \epsilon, \qquad \|f - s\|_q < \epsilon.$$

Moreover, choosing in the proof of Fact B.21 a continuous g with

$$\mu(\{x : g(x) \neq s(x)\}) < \min\Big\{\Big(\frac{\epsilon}{2\|s\|_\infty}\Big)^p, \Big(\frac{\epsilon}{2\|s\|_\infty}\Big)^q\Big\}, \qquad |g| \leq \|s\|_\infty,$$

we get simultaneously

$$\|f - g\|_p < 2\epsilon, \qquad \|f - g\|_q < 2\epsilon. \tag{20}$$

Examining the proof of Fact B.22 in the same way, for every $f \in L^p(\mathbb{R}) \cap L^q(\mathbb{R})$ there is $g \in C_c(\mathbb{R})$ with (20). Hence if $f \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$ there is a sequence $\langle g_n \rangle$ in $C_c(\mathbb{R})$ with

$$\|f - g_n\|_1 \to 0, \qquad \|f - g_n\|_2 \to 0,$$

and then

$$\lim_{n\to\infty} \|\widehat{f} - \widehat{g_n}\|_\infty = \lim_{n\to\infty} \|\mathcal{F}(f) - \mathcal{F}(g_n)\|_2 = 0.$$

Since $\widehat{g_n} = \mathcal{F}(g_n)$, we conclude $\widehat{f} = \mathcal{F}(f)$ (Exercise 6.3.4).

Exercise 6.3.4

If a sequence $\langle h_n \rangle$ in $C_0(\mathbb{R})$ satisfies $\lim_n \|f - h_n\|_\infty = \lim_n \|g - h_n\|_2 = 0$, show that $f = g$ almost everywhere.

Solution [Supplement]**.** Since $\|g - h_n\|_2 \to 0$, by Fact B.18 a subsequence $h_{n_k} \to g$ almost everywhere. But $h_{n_k} \to f$ uniformly, in particular pointwise everywhere. Hence $f = g$ almost everywhere. □

Surjectivity. By the proof of Theorem 5.2.8, every $h \in C_c^2(\mathbb{R})$ is the Fourier transform of a bounded L^1-function k ($k = \frac{1}{2\pi}\widehat{h}(-\cdot)$, which is bounded and integrable). A bounded L^1-function is an L^2-function, so $h = \widehat{k} = \mathcal{F}(k)$ lies in the range of $\mathcal{F}$. Hence the range of $\mathcal{F}$ is dense in $L^2(\widehat{\mathbb{R}})$; being the isometric image of a complete space it is closed, so $\mathcal{F} : L^2(\mathbb{R}) \to L^2(\widehat{\mathbb{R}})$ is a surjective isometry, the **Plancherel transform**. An isometry between Hilbert spaces is of course a Hilbert space isomorphism (polarization). Summarizing:

Theorem 6.3.8 (Plancherel)

There is a unique Hilbert space isomorphism $\mathcal{F} : L^2(\mathbb{R}) \to L^2(\widehat{\mathbb{R}})$ with $\mathcal{F}(f) = \widehat{f}$ for every $f \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$.

(Uniqueness: $\mathcal{F}$ is determined on the dense subspace $C_c(\mathbb{R}) \subset L^1 \cap L^2$ and is continuous.)

If $f \in L^2(\mathbb{R})$, then $f_A = \chi_{[-A,A]} f \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$ and

$$\widehat{f_A}(\alpha) = \int_{-A}^{A} f(x) e^{-i\alpha x}\, dx.$$

Since $\lim_{A\to\infty} \|f - f_A\|_2 = 0$,

$$\lim_{A\to\infty} \|\mathcal{F}(f) - \widehat{f_A}\|_2 = 0,$$

the analogue of (18) for Fourier series. Finally, the statement that the Plancherel transform is a Hilbert space isomorphism reads

$$\int_{\mathbb{R}} f(x)\overline{g(x)}\, dx = \frac{1}{2\pi} \int_{\mathbb{R}} \mathcal{F}f(\alpha)\overline{\mathcal{F}g(\alpha)}\, d\alpha,$$

also called **Parseval's identity**.

Example 2

Applying Parseval to $f(x) = g(x) = e^{-|x|}$ (with $\widehat{f}(\alpha) = \frac{2}{1+\alpha^2}$),

$$\int_{-\infty}^{\infty} (e^{-|x|})^2\, dx = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\frac{2}{1+\alpha^2}\right]^2 d\alpha,$$

so

$$\int_{-\infty}^{\infty} \frac{1}{(1+\alpha^2)^2}\, d\alpha = \pi \int_0^{\infty} e^{-2x}\, dx = \frac{\pi}{2}.$$

Exercise 6.3.5

Compute $\displaystyle\int_{-\infty}^{\infty} \frac{\sin a\alpha \, \sin b\alpha}{\alpha^2}\, d\alpha$ for $a, b > 0$.

Solution [Supplement]**.** By Example 1 of §5.2, $\widehat{\chi_{[-a,a]}}(\alpha) = \frac{2\sin a\alpha}{\alpha}$, real-valued. Parseval with $f = \chi_{[-a,a]}$, $g = \chi_{[-b,b]}$ (both in $L^1 \cap L^2$):

$$\int_{\mathbb{R}} \chi_{[-a,a]}\chi_{[-b,b]}\, dx = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{2\sin a\alpha}{\alpha} \cdot \frac{2\sin b\alpha}{\alpha}\, d\alpha.$$

The left side is $2\min\{a, b\}$. Therefore

$$\int_{-\infty}^{\infty} \frac{\sin a\alpha \, \sin b\alpha}{\alpha^2}\, d\alpha = \pi \min\{a, b\}.$$

(For $a = b$ this gives the classical $\int_{\mathbb{R}} \frac{\sin^2 a\alpha}{\alpha^2}\, d\alpha = \pi a$.) □